

ESP32-P4

Espressif Product Packaging Information



ESPRESSIF

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This document summarizes the packaging requirements of Espressif's ESP32-P4 series of products, including the product silk marking, dry-packing requirements, and product packing.

1 ESP32-P4 Chip Packaging Information

This document summarizes the packaging requirements of Espressif's ESP32-P4 series of chips, including the chip silk marking, dry-packing requirements, and chip product packing.

1.1 Chip Silk Marking

Espressif chip silk marking provides information such as chip name, flash size, and operating temperature.

Marking Convention

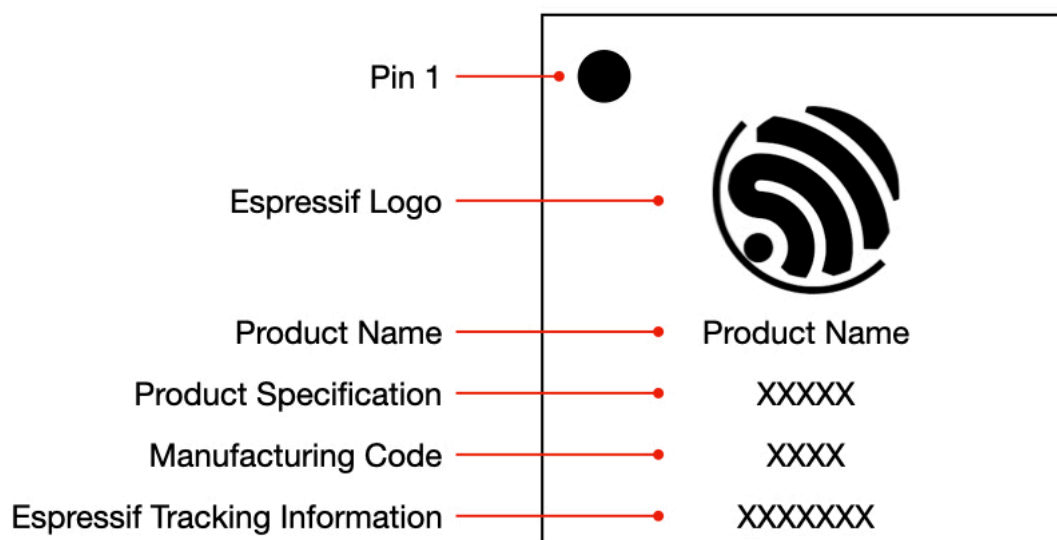


Fig. 1: Chip Silk Marking Diagram

- **Pin 1:** Position of Pin 1.
- **Espressif Logo:** Company logo.
- **Product Name:** Specifies the chip name.
- **Product Specification** is a code that indicates:
 - Operating temperature:**
 - * N = Up to 85 °C
 - * H = Up to 105 °C
 - (Optional) Flash and PSRAM configuration:**
 - * F = In-package flash
 - * R = In-package PSRAM
 - (Optional) Voltage level and interface:**
 - * No letter = 3.3 V Quad SPI
 - * V = 1.8 V Octal SPI

* W = 1.8 V Hex SPI

– (Optional) Flash or PSRAM size.

Examples:

–NRW16:

The chip supports up to 85 °C operating temperature, and includes in-package 1.8 V Hex SPI PSRAM (16 MB).

–NR4:

The chip supports up to 85 °C operating temperature, and includes in-package 3.3 V Quad SPI PSRAM (4 MB).

- **Manufacturing Code:** Identifies the chip revision.
- **Espressif Tracking Information:** Identifies the chip lot number.

1.2 Chip Packing

Espressif chips are packed on tape and reel. Then the reel is packed in an aluminum moisture barrier bag (MBB) in vacuum state to protect chips from absorbing moisture during transportation and storage. At last, the MBB is packed into a pizza box.

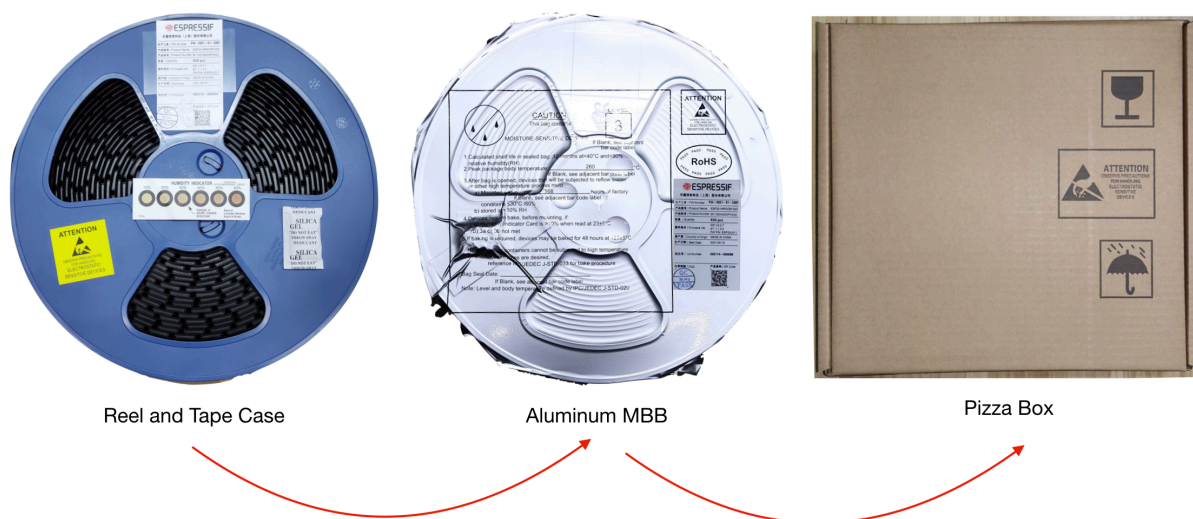


Fig. 2: Espressif Product Packing Method

Note: The figure(s) above is for illustration purposes only. Actual product may vary.

Tape

This section introduces the dimensions of the carrier tape.

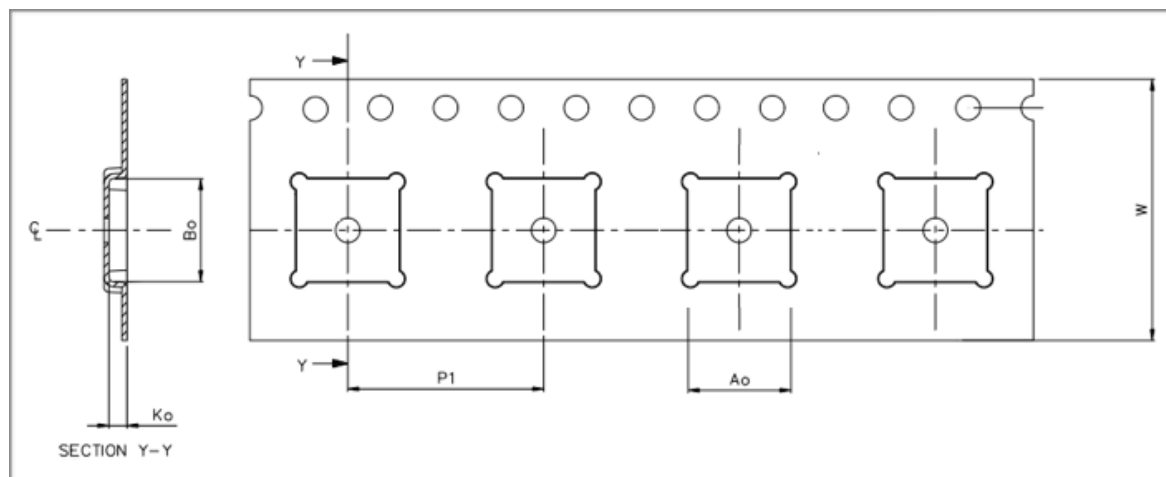


Table 1: Carrier Tape Dimensions (Unit: mm)

Package	Carrier Width (W)	Cavity Pitch (P_1)	Cavity Width (A_o)	Cavity Length (B_o)
10 * 10	24.0 ± 0.30	16.0 ± 0.10	10.30 ± 0.10	10.30 ± 0.10

Note: The surface resistance of the carrier tape is $10^4 \sim 10^{11}$ ohms.

Reel

This section introduces the dimensions of the reel.

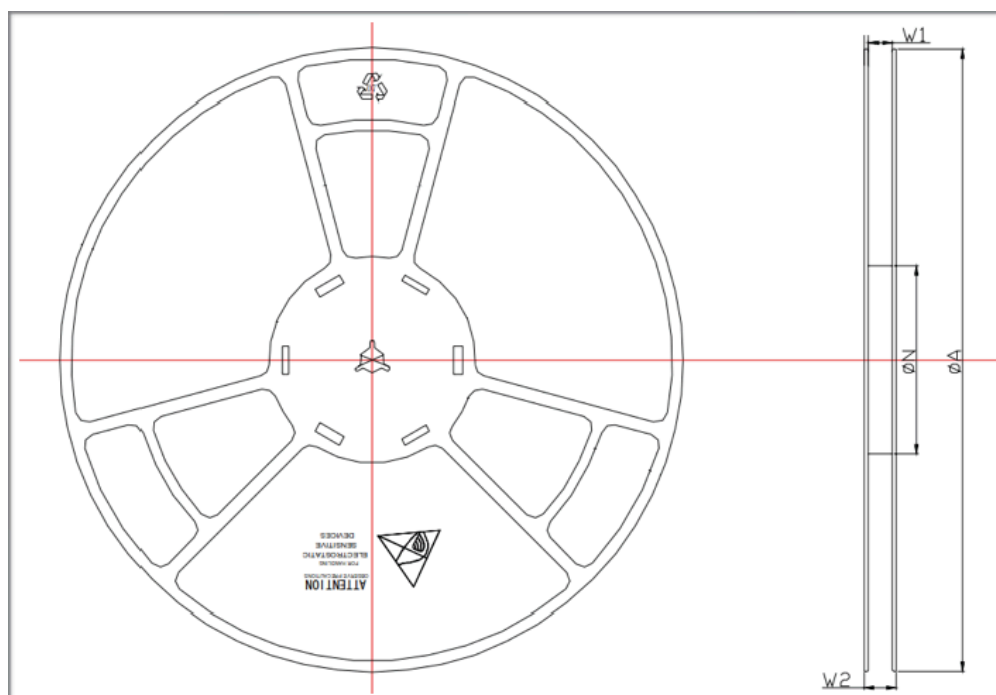


Table 2: Reel Dimensions

Package	Reel Size	Quantity Per Reel
10 * 10	13 ' '	2,000

Note: The surface resistance of the carrier tape is $10^4 \sim 10^{11}$ ohms.

Pin1 Location

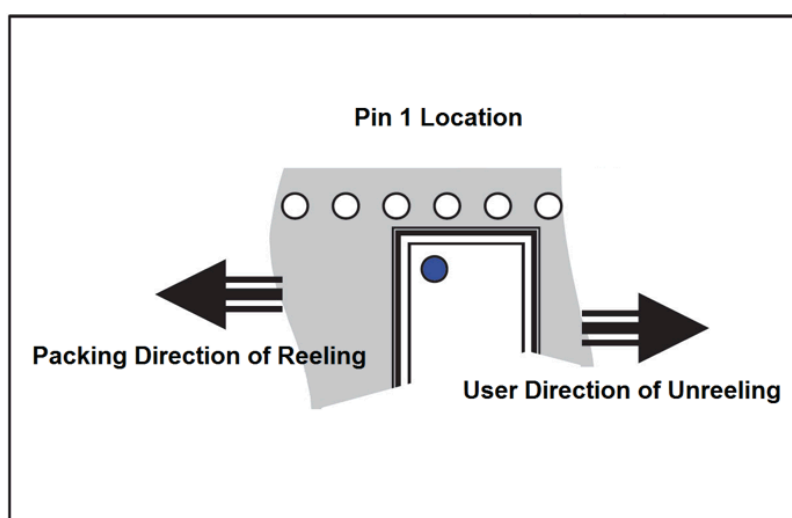


Fig. 3: Pin 1 Orientation of Chips in Carrier Tape

Pizza Box

Inside of the pizza box of typical Espressif chips, together with the tape and reel, product label and dry-packing related items are also packed.

This section describes the product label. For information about dry packing related items, please go to Section [Dry Packing Requirement](#).

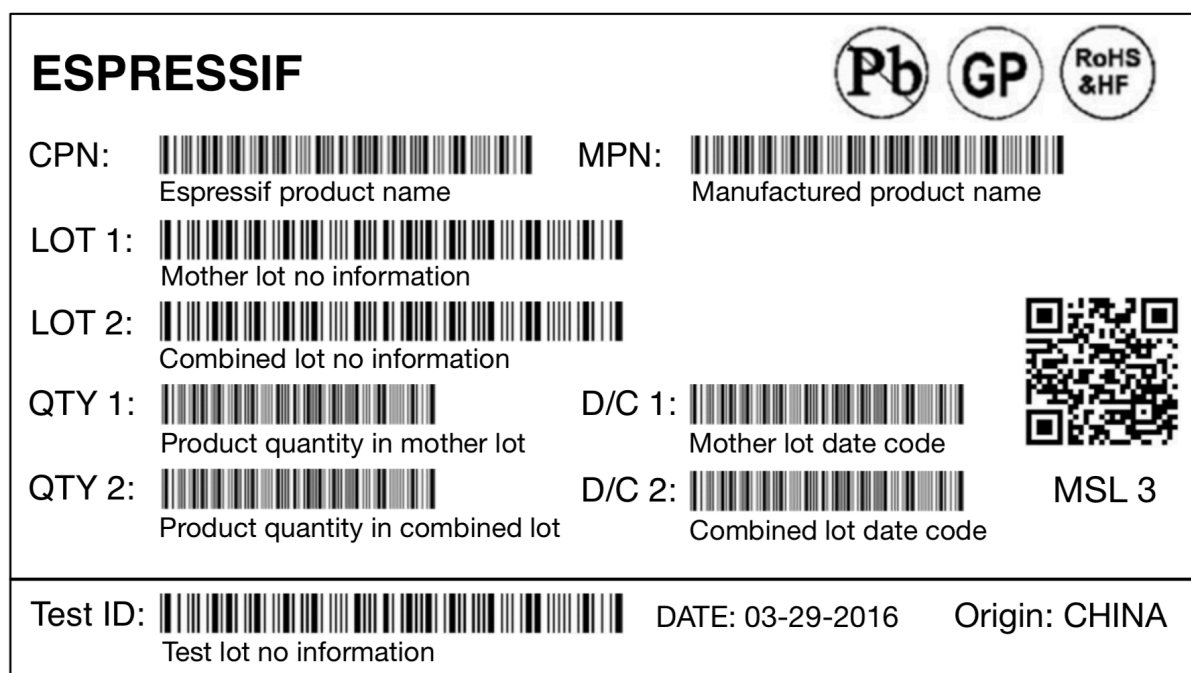


Fig. 4: Espressif Chip Label - One

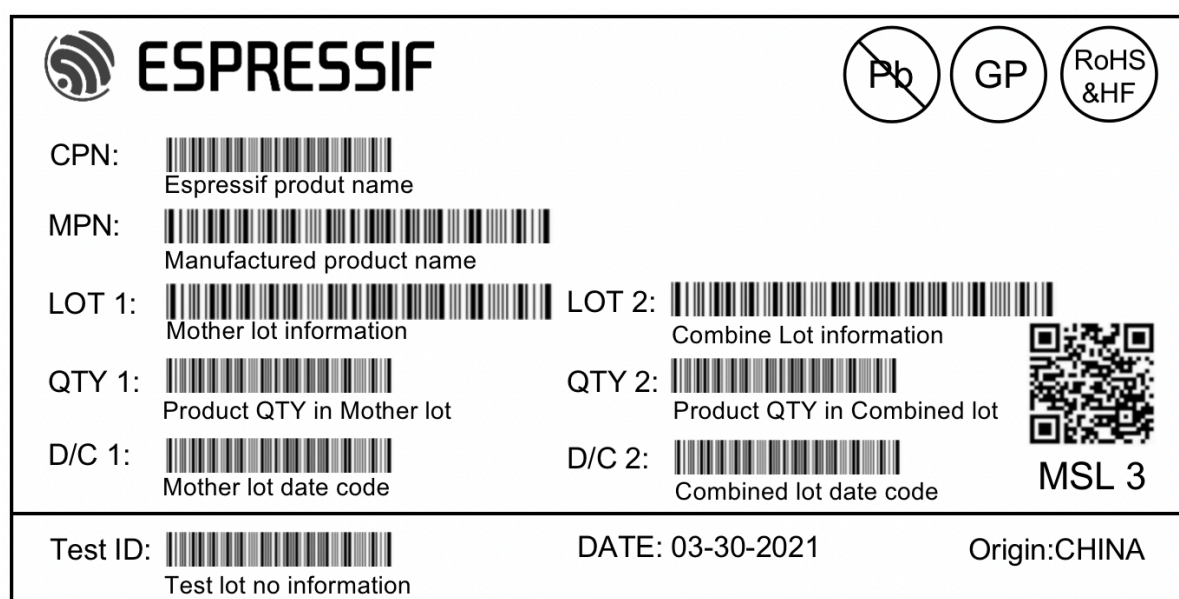


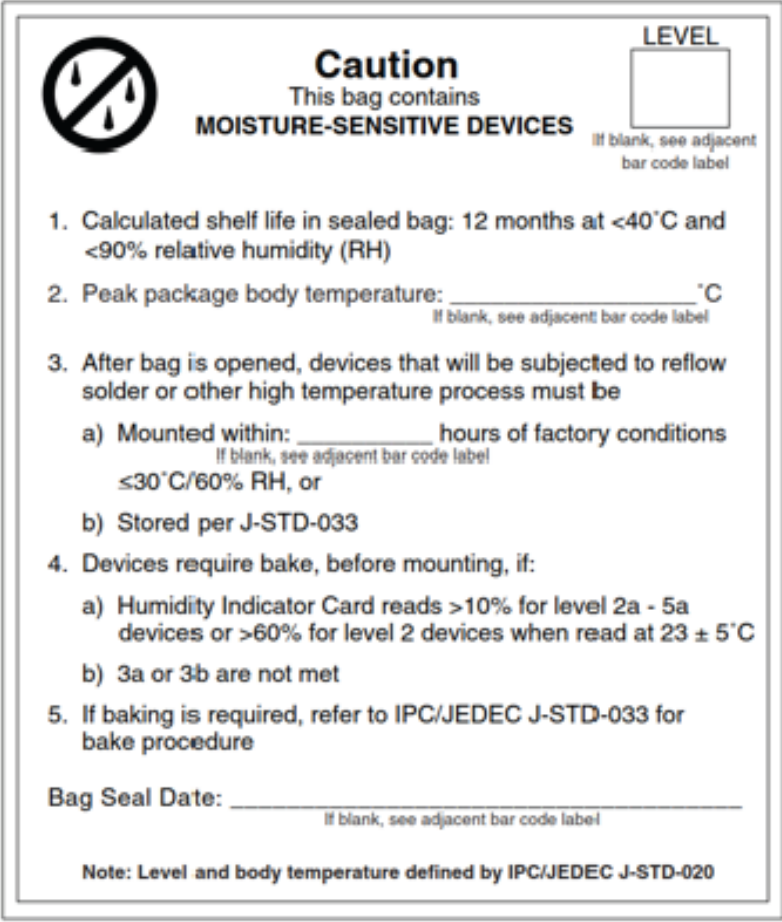
Fig. 5: Espressif Chip Label - Two

- **ESPRESSIF logo:** Company logo and name
- **CPN:** Espressif product name, e.g. ESP32-D0WDQ6
- **MPN:** Manufacturing product name
- **LOT1:** Number of mother lot
- **LOT2:** Number of combined lot
- **QTY1:** Quantity of mother lot
- **QTY2:** Quantity of combined lot
- **D/C 1:** Assembly date code for LOT1
- **D/C 2:** Assembly date code for LOT2

- **Test ID:** Test information
- **DATE:** Packing date, MM-DD-YYYY, DATE “03-29-2016” stands for Mar 29th in 2016

1.3 Dry Packing Requirement

All Espressif chip's moisture sensitivity level (MSL) is 3, thus must be dry-packed. Dry packing usually consists of desiccant material, a humidity-indicator card (HIC), as well as a Moisture Sensitivity Caution Label sealed with the populated reel inside a moisture barrier bag (MBB).



Caution
This bag contains
MOISTURE-SENSITIVE DEVICES

LEVEL
If blank, see adjacent bar code label

1. Calculated shelf life in sealed bag: 12 months at <40°C and <90% relative humidity (RH)
2. Peak package body temperature: _____ °C
If blank, see adjacent bar code label
3. After bag is opened, devices that will be subjected to reflow solder or other high temperature process must be
 - a) Mounted within: _____ hours of factory conditions
If blank, see adjacent bar code label
≤30°C/60% RH, or
 - b) Stored per J-STD-033
4. Devices require bake, before mounting, if:
 - a) Humidity Indicator Card reads >10% for level 2a - 5a devices or >60% for level 2 devices when read at 23 ± 5°C
 - b) 3a or 3b are not met
5. If baking is required, refer to IPC/JEDEC J-STD-033 for bake procedure

Bag Seal Date: _____
If blank, see adjacent bar code label

Note: Level and body temperature defined by IPC/JEDEC J-STD-020

Fig. 6: Moisture Sensitivity Caution Label

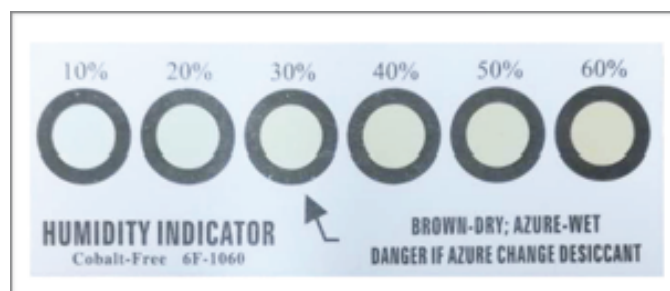


Fig. 7: Humidity-indicator Card

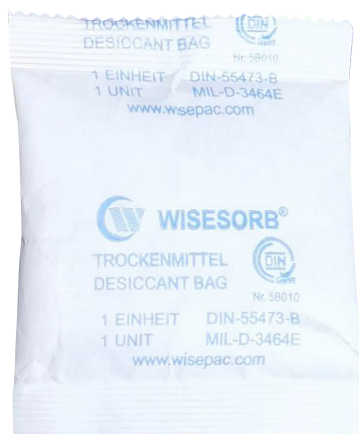


Fig. 8: Desiccant

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The floor life (the allowable period of time, after removal from a moisture barrier bag, dry storage or dry bake and before the reflow soldering process) is shown in the table below.

Level	Floor life (out of bag) at factory ambient $\leq 30^{\circ}\text{C}/60\% \text{ RH}$, or as stated
3	168 hours

Products require bake, before mounting, if:

- The humidity-indicator card reads $> 10\%$, when reading at $23 \pm 5^{\circ}\text{C}$;
- Or the period of time after removal from a moisture barrier bag or dry storage or dry bake and before the reflow soldering process is larger than the value listed in table above.

If baking is required, make sure that the products are taken out of the tape, and IPC/JEDEC J-STD-033 is followed during the bake procedure.

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